

AIM: Convert the ER diagram into a relational Schema. Write complete MySQL CREATE TABLE statements with primary key, foreign key, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert Sample data and demonstrate referential integrity violated.

SOURCE CODE:

Table	Primary key	Foreign Key
Customer	Customer-ID	
Customer_Phone	(Customer ID, Phone)	Customer ID - Customer
ADDRESS	ADDRESS ID	
SELLER	SELLER ID	Customer ID - Customer
Category	Category ID	-
Product	Product ID	-
Product-SELLER	(ProductID, SELLER-ID)	Category ID - Category
ELECTRONICS	Product ID	Product ID - Product
CLOTHING	Product ID	Seller ID - Seller
GROCERY	Product ID	Product ID - Product
ORDER	ORDERID	Product ID - Product
ORDER-Item	(ORDER ID, PRODUCTID)	Product ID - Product
PAYMENT	PAYMENT-ID	Customer ID - Customer
DELIVERY	DELIVERY-ID	Order ID - Order
		Order ID - Orders
		Order ID - orders

SQL Queries

→ Schema Creation

```
CREATE DATABASE E-COMMERCE-ER;
USE E-COMMERCE-ER;
CREATE TABLE Customer (
    Customer ID INT Primary Key,
    F-name VARCHAR(30) NOT NULL,
    L-name VARCHAR(30) NOT NULL,
    Email VARCHAR(100) NOT NULL,
    Date-Of-Birth DATE NOT NULL,
);
```

```
CREATE TABLE Customer - PHONE (
    CustomerID INT,
    Phone VARCHAR (15),
    Primary Key (Customer ID, PHONE);
    Foreign Key (Customer ID) References
    Customer (Customer ID) ON DELETE
    CASCADE
);
```

```
CREATE TABLE DATABASE (
    Address ID INT Primary Key,
    House - No. VARCHAR (10),
    Street VARCHAR (50),
    City VARCHAR (30) NOT NULL,
    PINCODE VARCHAR (20) NOT NULL,
    Address - Type VARCHAR (20),
    Customer - ID INT NOT NULL,
    FOREIGN KEY (Customer ID) References
    Customer (Customer ID) ON DELETE CASCADE
);
```

```
CREATE TABLE CATEGORY (
    Category ID INT PRIMARY KEY,
    Category - Name VARCHAR (50) NOT NULL UNIQUE,
    DESCRIPTION VARCHAR (150)
);
```

```
CREATE TABLE PRODUCT (
    Product ID INT Primary Key,
    Product - Name VARCHAR (100) NOT NULL,
    Price DECIMAL [10, 2] NOT NULL,
    Description VARCHAR (200),
    Stock Quantity INT NOT NULL Default 0,
    Brand VARCHAR,
    Category ID INT,
    FOREIGN Key (Category ID) References
    Category (Category ID) ON DELETE SET NULL
);
```

CREATE TABLE PRODUCT-SELLER (

Product ID INT,

Seller ID INT,

Primary Key (Product ID, Seller ID);

Foreign Key (Product ID) References

Product (Product ID) ON DELETE CASCADE

FOREIGN KEY (Seller ID) references (Seller ID) ON DELETE CASCADE
);

CREATE TABLE ELECTRONICS (

Product ID INT PRIMARY KEY,

Warranty-Period VARCHAR (50),

Model-Number VARCHAR (50);

FOREIGN KEY (Product ID) References Product (Product ID)
ON DELETE CASCADE
);

CREATE TABLE ORDERS (

ORDER INT PRIMARY KEY,

ORDER-DATE DATE NOT NULL,

ORDER-STATUS VARCHAR (20) NOT NULL,

Total Amount Decimal (10,2) NOT NULL,

Customer ID INT NOT NULL,

Foreign Key (Customer ID) REFERENCES Customer (Customer ID)
ON DELETE CASCADE
);

CREATE TABLE PAYMENTS (

Payment-ID INT PRIMARY KEY,

Payment-DATE DATE NOT NULL,

Amount DECIMAL (10,2) NOT NULL,

Payment-METHOD VARCHAR (30) NOT NULL,

Payment-STATUS VARCHAR (30) NOT NULL,

Transaction ID VARCHAR (30) UNIQUE,

FOREIGN KEY (ORDER ID) References Orders (Order ID) ON DELETE CASCADE
);

CREATE TABLE DELIVERY (

~~Payment-ID~~ Delivery ID INT PRIMARY KEY,

Delivery-Date DATE,

Delivery-Status VARCHAR (20) NOT NULL,

Tracking-ID VARCHAR (30) UNIQUE,

Delivery-Partner VARCHAR (50),

Order ID INT NOT NULL UNIQUE,

FOREIGN KEY (ORDER ID) References Orders (Order ID) ON DELETE CASCADE
);

USE ECOMMERCE - ER;

INSERT INTO Customer VALUES

(1, 'Ishaan', 'Sharma', 'SharmaIshaan@gmail.com', '2006-15-09');

INSERT INTO CUSTOMER - PHONE VALUES

(1, '9919101807');

(1, '9838302038');

INSERT INTO ADDRESS VALUES

(4, '307', 'UNNAO', 'LUCKNOW', 'UP', '209859', 'College', 1);

INSERT INTO SELLER VALUES

(101, 'TechStore', 'GST1234567');

INSERT INTO CATEGORY VALUES

(201, 'Electronics', 'Electronic gadgets and accessories');

INSERT INTO PRODUCT VALUES

(301, 'Gaming Laptop', '75,000.00', '15.6 INCH', 'RTX5080', 10, 'ASUS', 20);

INSERT INTO ELECTRONICS VALUES

(301, '2 years', 'ASUS-TUFF-FIS');

INSERT INTO PRODUCT-SELLER VALUES

(301, 101);

INSERT INTO ORDERS VALUES

(501, '2026-08-14', 'Placed', \$75000.00);

INSERT INTO ORDERITEM VALUES

(501, 301, 75000.00, 75000.00);

INSERT INTO PAYMENT VALUES

(601, '2026-08-14', 75000.00, 'UPI', 'PAID', 'TXN123', 1501);

INSERT INTO DELIVERY VALUES

(701, NULL, 'PROCESSING', 'TRX123', 'Delivery', 501);

→ Referential Integrity Violation

Insert INTO ORERS VALUES

(502, '2026-08-14', 'Placed', \$50000.00, 999);

→ ON DELETE SET NULL Demonstration

DELETE FROM CATEGORY WHERE CATEGORY

(0 = 210);

→ OUTPUT - SHOW TABLES

Show Tables;

Tables - IN - E - COMMERCE - ER

ADDRESS
CATEGORY
CLOTHING
CUSTOMER
CUSTOMER - PHONE
DELIVERY
ELECTRONICS
ORDER ITEM
GROCERY
ORDERS
PAYMENT
PRODUCT
PRODUCT - SELLER
SELLER

OUTPUT-2:

Select * FROM CUSTOMER;

Customer ID	Fname	Lname	Email	Date of Birth
1	Ishaan	Sharma	SharmaIshaan@gmail.com	2006-15-09

Select * FROM PRODUCT;

Product ID	Product Name	Price	Stock QTY	Brand	Catg. ID
301	Laptop	75000.00	10	ASUS	201

Select * FROM Orders;

Order ID	Order Date	Order Status	Total Amount	Customer ID
501	2026-08-14	Placed	75000.00	1

OUTPUT-3: Referential Integrity Violation

- Since Customer ID = 999 does not exist in Customer;
(ERROR 1452/23000): cannot add or update a child row:
a foreign key constraint fails ('E-COMMERCE-ER'), ('ORDERS'), ('CONSTRAINTS'),
('ORDERS-IEFK-1') FOREIGN KEY (Customer ID)
References 'Customer' ('Customer ID') ON DELETE CASCADE

Reason: Customer ID 999 does not exist in Customer

Before Deletion:

Product ID
301

Product - Name
Laptop

Category ID
201

DELETE FROM CATEGORY WHERE CATEGORY ID = 201;

After Deletion:

Product ID
301

Product - Name
Laptop

Category ID
NULL

OUTPUT - 5: ON DELETE CASCADE (Customer → everything downstream)

- Before deleting Customer ID = 1, related rows exist in ORDERS, ORDERITEM, PAYMENT, DELIVERY, ADDRESS and CUSTOMER_PHONE

(1 row each / 1 order item).

Delete From Customer where Customer ID = 1;

Query OK, 1 row affected.

TABLE

ROW COUNT AFTER DELETE

ORDERS

0

ORDERITEM

0

PAYMENT

0

DELIVERY

0

ADDRESS

0

CUSTOMER_PHONE

0